AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Course Title

INTRODUCTION TO DATABASE (2108)

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Section: [F]

TITLE

Gym Management System

Supervised By

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Introduction

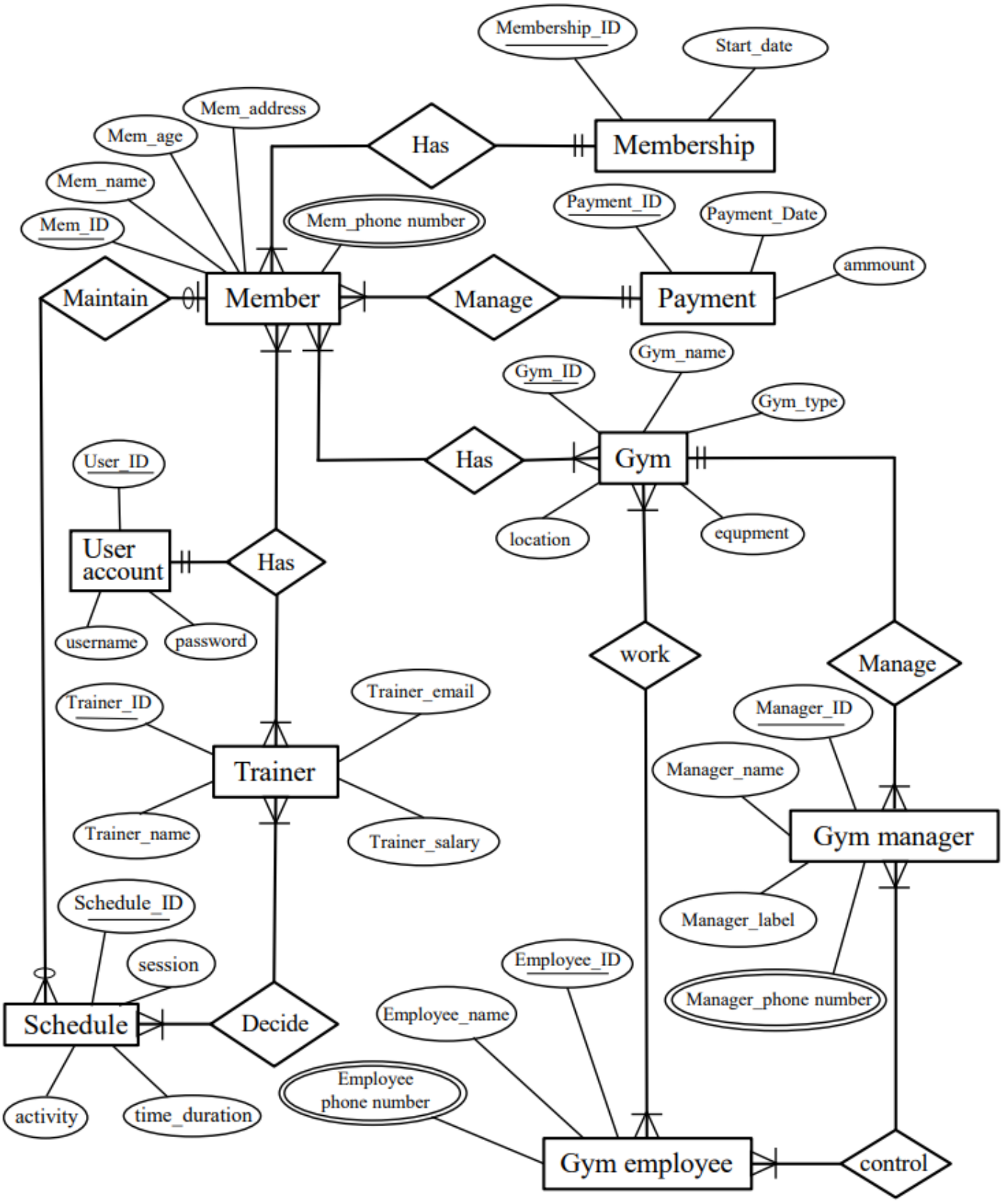
Our gym management system is a gym and health membership management system. Here, we can keep records of our members and their membership. Gym management also includes payment, trainers, schedules, and other reports that help the management system. It is a complete gym and recreation facility system that looks after all the members, membership, and activities. Our gym management system provides lots of functions, such as a user account that keeps records of all of the things about payment, schedule, and trainer. Which provide good quality services to members from gym managers and employees. It's all about our gym management system.

Case Study / Scenario

StudentID1: 22-49827-3 Name: Arpita Barua	StudentID3: 23-50170-1 Name: Fatima
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CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

The gym management system is designed to streamline operations and enhance the overall experience for gym members, employees, and managers. Every information of the gym will be stored in the database. There will be a gym, and it will have location, name, gym type, and equipment, which will be stored in the database. Gyms are identified by gym ID. Many members can join the gym. Each member is assigned a unique member ID for identification. The database records the the name, age, phone number, and address of the member. All members have membership. There is a unique membership ID and start date. Each member can have exactly one membership, and each membership has many members. Each member will have exactly one user account with their unique user ID, username, and password. Member manages payment. Payment details such as a unique payment ID, payment date, and amount. Payment represents the payment transactions for the member. Each member can have multiple payment transactions. The gym manager of the gym will manage the whole system of the gym. A gym may have many gym managers, or a gym manager can manage the gym. Every gym manager is identified by his manager ID. They have also a name, label and phone number. Gym employees represent the employees working at the gym. Gym managers control multiple gym employees, and each employee can work at one gym. Every employee has a unique employee ID, name, and phone number. The trainer has responsibility for providing training sessions to members. Members are trained by many trainers. Every trainer is identified by a trainer ID. Every trainer also has a name, salary, and email. The trainer decides the schedule for members. Each trainer can have multiple schedules. Members maintain their own schedule. Schedule is identified by it's schedule ID. The schedule has sessions, time durations and activities.

ER Diagram


Membership_ID
Start_date
Mem_address
Mem_age
Has
Membership
Mem_name
Mem_ID
Mem_phone number
Payment_ID
Payment_Date
ammount
Maintain
Member
Manage
Payment
Gym_name
Gym_ID
Gym_type
User_ID
Has
Gym
location
equpment
User account
Has
username
password
work
Manage
Trainer_ID
Trainer_email
Manager_ID
Trainer
Manager_name
Trainer_name
Trainer_salary
Gym manager
Schedule_ID
Manager_label
session
Employee_ID
Manager_phone number
Schedule
Decide
Employee_name
Employee phone number
activity
time_duration
Gym employee
control

Normalization

Has (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID, Start_date)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID

Membership_ID, Start_date

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID

Membership_ID, Start_date

No Transitive Dependency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID

Membership_ID, Start_date

Manage (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID, Payment_date, ammount)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID

Payment_ID, Payment_date, amount

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID

Payment_ID, Payment_date, amount

No Transitive Dependency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID

Payment_ID, Payment_date, amount

Has (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Gym_ID, Gym_name, Gym_type, location, equipment)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Gym_ID, Gym_name, Gym_type, location, equipment

MG_ID, Mem_ID, Gym_ID

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Gym_ID, Gym_name, Gym_type, location, equipment

MG_ID, Mem_ID, Gym_ID

No Transitive Dependency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Gym_ID, Gym_name, Gym_type, location, equipment

MG_ID, Mem_ID, Gym_ID

Work (Gym_ID, Gym_name, Gym_type, location, equipment, Employee_ID, Employee_name, Employee_phone number)

1NF: Employee_phone number Multivalued Attribute

2NF: Gym_ID, Gym_name, Gym_type, location, equipment

Employee_ID, Employee_name, Employee_phone number

GE_ID, Gym_ID, Employee_ID

3NF: Gym_ID, Gym_name, Gym_type, location, equipment

Employee_ID, Employee_name, Employee_phone number

GE_ID, Gym_ID, Employee_ID

No Transitive Dependency

Table: Gym_ID, Gym_name, Gym_type, location, equipment

Employee_ID, Employee_name, Employee_phone number

GE_ID, Gym_ID, Employee_ID

Control (Manager_ID, Manager_name, Manager_label, Manager_phone number, Employee_ID, Employee_name, Employee_phone number)

1NF: Manager_phone number, Employee_phone number Multivalued Attribute

2NF: Manager_ID, Manager_name, Manager_label, Manager_phone number

Employee_ID, Employee_name, Employee_phone number

ME_ID, Manager_ID, Employee_ID

3NF: Manager_ID, Manager_name, Manager_label, Manager_phone number

Employee_ID, Employee_name, Employee_phone number

ME_ID, Manager_ID, Employee_ID

No Transitive Dependency

Table: Manager_ID, Manager_name, Manager_label, Manager_phone number

Employee_ID, Employee_name, Employee_phone number

ME_ID, Manager_ID, Employee_ID

Manage (Gym_ID, Gym_name, Gym_type, location, equipment, Manager_ID, Manager_name, Manager_label, Manager_phone number)

1NF: Manager_phone number Multivalued Attribute

2NF: Gym_ID, Gym_name, Gym_type, location, equipment

Manager_ID, Manager_name, Manager_label, Manager_phone number, Gym_ID

3NF: Gym_ID, Gym_name, Gym_type, location, equipment

Manager_ID, Manager_name, Manager_label, Manager_phone number, Gym_ID

No Transitive Dependency

Table: Gym_ID, Gym_name, Gym_type, location, equipment

Manager_ID, Manager_name, Manager_label, Manager_phone number, Gym_ID

Decide (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary, Schedule_ID, session, activity, time_duration)

1NF: No Multivalued Attribute

2NF: Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

Schedule_ID, session, activity, time_duration

TS_ID, Trainer_ID, Schedule_ID

3NF: Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

Schedule_ID, session, activity, time_duration

TS_ID, Trainer_ID, Schedule_ID

No Transitive Dependency

Table: Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

Schedule_ID, session, activity, time_duration

TS_ID, Trainer_ID, Schedule_ID

Has (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Trainer_ID, Trainer_name, Trainer_email, Trainer_salary)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

MT_ID, Mem_ID, Trainer_ID

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

MT_ID, Mem_ID, Trainer_ID

No Transitive Depenedency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Trainer_ID, Trainer_name, Trainer_email, Trainer_salary

MT_ID, Mem_ID, Trainer_ID

Has (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID, Username, Password)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID

User_ID, Username, Password

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID

User_ID, Username, Password

No Transitive Dependency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID

User_ID, Username, Password

Maintain (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Schedule_ID, session, activity, time_duration)

1NF: Mem_phone number Multivalued Attribute

2NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Schedule_ID, session, activity, time_duration, Mem_ID

3NF: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Schedule_ID, session, activity, time_duration, Mem_ID

No Transitive Dependency

Table: Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number

Schedule_ID, session, activity, time_duration, Mem_ID

Finalization

Total Table:

1. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID √
2. Membership_ID, Start_date √
3. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID √
4. Payment_ID, Payment_date, amount √
5. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number ×
6. Gym_ID, Gym_name, Gym_type, location, equipment √
7. MG_ID, Mem_ID, Gym_ID √
8. Gym_ID, Gym_name, Gym_type, location, equipment ×
9. Employee_ID, Employee_name, Employee_phone number √
10. GE_ID, Gym_ID, Employee_ID √
11. Manager_ID, Manager_name, Manager_label, Manager_phone number ×
12. Employee_ID, Employee_name, Employee_phone number ×
13. ME_ID, Manager_ID, Employee_ID √
14. Gym_ID, Gym_name, Gym_type, location, equipment ×
15. Manager_ID, Manager_name, Manager_label, Manager_phone number, Gym_ID √
16. Trainer_ID, Trainer_name, Trainer_email, Trainer_salary √
17. Schedule_ID, session, activity, time_duration ×
18. TS_ID, Trainer_ID, Schedule_ID √
19. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number ×
20. Trainer_ID, Trainer_name, Trainer_email, Trainer_salary ×
21. MT_ID, Mem_ID, Trainer_ID √
22. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID √
23. User_ID, Username, Password √

24. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number ×

25. Schedule_ID, session, activity, time_duration, Mem_ID √

Final Table:

1. Membership_ID, Start_date (Membership)
2. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Membership_ID (Member)
3. Payment_ID, Payment_date, amount (Payment_Info)
4. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, Payment_ID (Members_Payment)
5. Gym_ID, Gym_name, Gym_type, location, equipment (Gym)
6. MG_ID, Mem_ID, Gym_ID (Member_associate_Gym)
7. Employee_ID, Employee_name, Employee_phone number (Gym_Employee)
8. GE_ID, Gym_ID, Employee_ID (Emp_In_Gym)
9. Manager_ID, Manager_name, Manager_label, Manager_phone number, Gym_ID (Manager)
10. ME_ID, Manager_ID, Employee_ID (Emp_and_Manager)
11. Trainer_ID, Trainer_name, Trainer_email, Trainer_salary (Trainer_info)
12. Schedule_ID, session, activity, time_duration, Mem_ID (Schedule)
13. TS_ID, Trainer_ID, Schedule_ID (Trainer_Period)
14. MT_ID, Mem_ID, Trainer_ID (Mem_and_Trainer)
15. User_ID, Username, Password (User_account_Info)
16. Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone number, User_ID (Mem_User_account)

Table Creation (DDL Operations)

StudentID1: 22-49827-3 Name: Arpita Barua	StudentID3: 23-50170-1 Name: Fatima
StudentID2: 23-50150-1 Name: Junayed Islam	StudentID4: 23-50683-1 Name: Ragib Nadim
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

1. Membership:

```
create table Membership
(
Membership_ID varchar2(10) constraint Membership_Membership_ID_pk primary key,
Start_date date
)
```

Object Type **TABLE** Object **MEMBERSHIP**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEMBERSHIP	MEMBERSHIP_ID	Varchar2	10	-	-	1	-	-	-
	START_DATE	Date	7	-	-	-	✓	-	-

1 - 2

2. Member:

```
create table Member
(
Mem_ID varchar2(10) constraint Member_Mem_ID_pk primary key,
Mem_name varchar2(50),
Mem_age number(3),
Mem_address varchar2(50),
Mem_phone_number varchar2(50),
Membership_ID varchar2(10) constraint Member_Membership_ID_fk references
Membership(Membership_ID)
)
```

Object Type **TABLE** Object **MEMBER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEMBER	MEM_ID	Varchar2	10	-	-	1	-	-	-
	MEM_NAME	Varchar2	50	-	-	-	✓	-	-
	MEM_AGE	Number	-	3	0	-	✓	-	-
	MEM_ADDRESS	Varchar2	50	-	-	-	✓	-	-
	MEM_PHONE_NUMBER	Varchar2	50	-	-	-	✓	-	-
	MEMBERSHIP_ID	Varchar2	10	-	-	-	✓	-	-

1 - 6

3. Payment_Info:

```
create table Payment_Info
(
Payment_ID varchar2(10) constraint Payment_Info_Payment_ID_pk primary key,
Payment_date date,
amount varchar2(10)
)
```

Object Type **TABLE** Object **PAYMENT_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT_INFO	PAYMENT_ID	Varchar2	10	-	-	1	-	-	-
	PAYMENT_DATE	Date	7	-	-	-	✓	-	-
	AMOUNT	Varchar2	10	-	-	-	✓	-	-
									1 - 3

4. Members_Payment:

```
create table Members_Payment
(
Mem_ID varchar2(10) constraint Members_Payment_Mem_ID_pk primary key,
Mem_name varchar2(50),
Mem_age number (3),
Mem_address varchar2(50),
Mem_phone_number varchar2(50),
Payment_ID varchar2(10) constraint Members_Payment_Payment_ID_fk references
Payment_Info(Payment_ID)
)
```

Object Type **TABLE** Object **MEMBERS_PAYMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEMBERS_PAYMENT	MEM_ID	Varchar2	10	-	-	1	-	-	-
	MEM_NAME	Varchar2	50	-	-	-	✓	-	-
	MEM_AGE	Number	-	3	0	-	✓	-	-
	MEM_ADDRESS	Varchar2	50	-	-	-	✓	-	-
	MEM_PHONE_NUMBER	Varchar2	50	-	-	-	✓	-	-
	PAYMENT_ID	Varchar2	10	-	-	-	✓	-	-
									1 - 6

5. Gym:

```
create table Gym
(
Gym_ID varchar2(10) constraint Gym_Gym_ID_pk primary key,
Gym_name varchar2(30),
Gym_type varchar2(30),
location varchar2(40),
equipment varchar2(20)
)
```

Object Type **TABLE** Object **GYM**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
GYM	GYM_ID	Varchar2	10	-	-	1	-	-	-
	GYM_NAME	Varchar2	30	-	-	-	✓	-	-
	GYM_TYPE	Varchar2	30	-	-	-	✓	-	-
	LOCATION	Varchar2	40	-	-	-	✓	-	-
	EQUIPMENT	Varchar2	20	-	-	-	✓	-	-
									1 - 5

6. Member_associate_Gym:

```
create table Member_associate_Gym
(
MG_ID varchar2(10) constraint Member_associate_Gym_MG_ID_pk primary key,
Mem_ID varchar2(10) constraint Member_associate_Gym_Mem_ID_fk references Member (Mem_ID),
Gym_ID varchar2(10) constraint Member_associate_Gym_Gym_ID_fk references Gym (Gym_ID)
)
```

Object Type **TABLE** Object **MEMBER_ASSOCIATE_GYM**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEMBER_ASSOCIATE_GYM	MG_ID	Varchar2	10	-	-	1	-	-	-
	MEM_ID	Varchar2	10	-	-	-	✓	-	-
	GYM_ID	Varchar2	10	-	-	-	✓	-	-
									1 - 3

7. Gym_Employee:

```
create table Gym_Employee
(
Employee_ID varchar2(10) constraint Gym_Employee_Employee_ID_pk primary key,
Employee_name varchar2(30),
Employee_phone_number varchar2(50) )
```

Object Type **TABLE** Object **GYM_EMPLOYEE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
GYM_EMPLOYEE	EMPLOYEE_ID	Varchar2	10	-	-	1	-	-	-
	EMPLOYEE_NAME	Varchar2	30	-	-	-	✓	-	-
	EMPLOYEE_PHONE_NUMBER	Varchar2	50	-	-	-	✓	-	-
									1 - 3

8. Emp_In_Gym:

```
create table Emp_In_Gym
(
GE_ID varchar2(10) constraint Emp_In_Gym_GE_ID_pk primary key,
Gym_ID varchar2(10) constraint Emp_In_Gym_Gym_ID_fk references Gym (Gym_ID),
Employee_ID varchar2(10) constraint Emp_In_Gym_Employee_ID_fk references
Gym_Employee(Employee_ID)
)
```

Object Type **TABLE** Object **EMP_IN_GYM**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP_IN_GYM	GE_ID	Varchar2	10	-	-	1	-	-	-
	GYM_ID	Varchar2	10	-	-	-	✓	-	-
	EMPLOYEE_ID	Varchar2	10	-	-	-	✓	-	-

1 - 3

9. Manager:

```
create table Manager
(
Manager_ID varchar2(10) constraint Manager_Manager_ID_pk primary key,
Manager_name varchar2(30),
Manager_label varchar2(5),
Manager_phone_number varchar2(50),
Gym_ID varchar2(10) constraint Manager_Gym_ID_fk references Gym(Gym_ID)
)
```

Object Type **TABLE** Object **MANAGER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MANAGER	MANAGER_ID	Varchar2	10	-	-	1	-	-	-
	MANAGER_NAME	Varchar2	30	-	-	-	✓	-	-
	MANAGER_LABEL	Varchar2	5	-	-	-	✓	-	-
	MANAGER_PHONE_NUMBER	Varchar2	50	-	-	-	✓	-	-
	GYM_ID	Varchar2	10	-	-	-	✓	-	-

1 - 5

10. Emp_and_Manager:

```
create table Emp_and_Manager
(
ME_ID varchar2(10) constraint Emp_and_Manager_ME_ID_pk primary key,
Manager_ID varchar2(10) constraint Emp_and_Manager_Manager_ID_fk references Manager
(Manager_ID),
Employee_ID varchar2(10) constraint Emp_and_Manager_Employee_ID_fk references
Gym_Employee(Employee_ID)
)
```

Object Type **TABLE** Object **EMP_AND_MANAGER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP_AND_MANAGER	ME_ID	Varchar2	10	-	-	1	-	-	-
	MANAGER_ID	Varchar2	10	-	-	-	✓	-	-
	EMPLOYEE_ID	Varchar2	10	-	-	-	✓	-	-

1 - 3

11. Trainer_Info:

```
create table Trainer_Info
(
Trainer_ID varchar2(10) constraint Trainer_Info_Trainer_ID_pk primary key,
Trainer_name varchar2(30),
Trainer_email varchar2(50),
Trainer_salary number(10)
)
```

Object Type **TABLE** Object **TRAINER_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TRAINER_INFO	TRAINER_ID	Varchar2	10	-	-	1	-	-	-
	TRAINER_NAME	Varchar2	30	-	-	-	✓	-	-
	TRAINER_EMAIL	Varchar2	50	-	-	-	✓	-	-
	TRAINER_SALARY	Number	-	10	0	-	✓	-	-

1 - 4

12. Schedule:

```
create table Schedule
(
Schedule_ID varchar2(10) constraint Schedule_Schedule_ID_pk primary key,
"Session" varchar2(20),
"Activity" varchar2(30),
Time_Duration varchar2(30),
Mem_ID varchar2(10) constraint Schedule_Mem_ID_fk references Member(Mem_ID)
)
```

Object Type **TABLE** Object **SCHEDULE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SCHEDULE	SCHEDULE_ID	Varchar2	10	-	-	1	-	-	-
	Session	Varchar2	20	-	-	-	✓	-	-
	Activity	Varchar2	30	-	-	-	✓	-	-
	TIME_DURATION	Varchar2	30	-	-	-	✓	-	-
	MEM_ID	Varchar2	10	-	-	-	✓	-	-

1 - 5

13. Trainer_Period:

```
create table Trainer_Period
(
TS_ID varchar2(10) constraint Trainer_Period_TS_ID_pk primary key,
Trainer_ID varchar2(10) constraint Trainer_Period_Trainer_ID_fk references Trainer_Info(Trainer_ID),
Schedule_ID varchar2(10) constraint Trainer_Period_Schedule_ID_fk references Schedule(Schedule_ID)
)
```

Object Type **TABLE** Object **TRAINER_PERIOD**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TRAINER_PERIOD	TS_ID	Varchar2	10	-	-	1	-	-	-
	TRAINER_ID	Varchar2	10	-	-	-	✓	-	-
	SCHEDULE_ID	Varchar2	10	-	-	-	✓	-	-

1 - 3

14. Mem_and_Trainer:

```
create table Mem_and_Trainer
(  MT_ID varchar2(10) constraint Mem_and_Trainer_MT_ID_pk primary key,
   Mem_ID varchar2(10) constraint Mem_and_Trainer_Mem_ID_fk references Member (Mem_ID),
   Trainer_ID varchar2(10) constraint Mem_and_Trainer_Trainer_ID_fk references
   Trainer_Info (Trainer_ID)  )
```

Object Type **TABLE** Object **MEM_AND_TRAINER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEM_AND_TRAINER	MT_ID	Varchar2	10	-	-	1	-	-	-
	MEM_ID	Varchar2	10	-	-	-	✓	-	-
	TRAINER_ID	Varchar2	10	-	-	-	✓	-	-

1 - 3

15. User_account_Info:

```
create table User_account_Info
(
User_ID varchar2(10) constraint User_account_User_ID_pk primary key,
Username varchar2(20),
Password varchar2(8)
)
```

Object Type **TABLE** Object **USER_ACCOUNT_INFO**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
USER_ACCOUNT_INFO	USER_ID	Varchar2	10	-	-	1	-	-	-
	USERNAME	Varchar2	20	-	-	-	✓	-	-
	PASSWORD	Varchar2	8	-	-	-	✓	-	-
									1 - 3

16. Mem_User_account:

```
create table Mem_User_account
(
Mem_ID varchar2(10) constraint Mem_User_account_Mem_ID_pk primary key,
Mem_name varchar2(50),
Mem_age number (3),
Mem_address varchar2(50),
Mem_phone_number varchar2(50),
User_ID constraint Mem_User_account_User_ID_fk references User_account_Info(User_ID)
)
```

Object Type **TABLE** Object **MEM_USER_ACCOUNT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MEM_USER_ACCOUNT	MEM_ID	Varchar2	10	-	-	1	-	-	-
	MEM_NAME	Varchar2	50	-	-	-	✓	-	-
	MEM_AGE	Number	-	3	0	-	✓	-	-
	MEM_ADDRESS	Varchar2	50	-	-	-	✓	-	-
	MEM_PHONE_NUMBER	Varchar2	50	-	-	-	✓	-	-
	USER_ID	Varchar2	10	-	-	-	✓	-	-
									1 - 6

Inserted Values in the tables

1. Membership:

insert into Membership (Membership_ID, Start_date) values ('24-01', '10-JAN-24')

insert into Membership (Membership_ID, Start_date) values ('24-02', '16-JAN-24')

insert into Membership (Membership_ID, Start_date) values ('24-03', '24-JAN-24')

insert into Membership (Membership_ID, Start_date) values ('24-04', '03-FEB-24')

insert into Membership (Membership_ID, Start_date) values ('24-05', '23-FEB-24')

insert into Membership (Membership_ID, Start_date) values ('24-06', '07-MAR-24')

insert into Membership (Membership_ID, Start_date) values ('24-07', '28-MAR-24')

insert into Membership (Membership_ID, Start_date) values ('24-08', '17-APR-24')

insert into Membership (Membership_ID, Start_date) values ('24-09', '17-APR-24')

insert into Membership (Membership_ID, Start_date) values ('24-10', '05-MAY-24')

MEMBERSHIP_ID	START_DATE
24-01	10-JAN-24
24-02	16-JAN-24
24-03	24-JAN-24
24-04	03-FEB-24
24-05	23-FEB-24
24-06	07-MAR-24
24-07	28-MAR-24
24-08	17-APR-24
24-09	17-APR-24
24-10	05-MAY-24

2. Member:

CREATE SEQUENCE Member_Mem_ID
INCREMENT BY 1
START WITH 101
MAXVALUE 99999
NOCACHE
NOCYCLE;
insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Abir', 25, 'Banani, Dhaka', '01998977150, 01705008327', '24-01')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Nadim', 23, 'Mirpur, Dhaka', '01925678942, 01735894254', '24-02')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Rasel', 26, 'Badda, Dhaka', '01316478594, 01716845972', '24-03')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Shuvro', 19, 'Uttara, Dhaka', '01648579582, 01912462573', '24-04')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Yeasir', 22, 'Gulshan, Dhaka', '01316489264, 01732485324', '24-05')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Junayed', 22, 'Bashundhara, Dhaka', '01998977150, 01705008327', '24-06')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Akash', 20, 'Mirpur, Dhaka', '01562146879, 01934586721', '24-07')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Abrar', 18, 'Banani, Dhaka', '01635453545, 01765789315', '24-08')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Shahriar', 27, 'Farmgate, Dhaka', '01993241567, 01701873598', '24-09')

insert into Member (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Membership_ID) values (Member_Mem_ID.nextval, 'Kabir', 21, 'Uttara, Dhaka', '01768352497, 01915482679', '24-10')

MEM_ID	MEM_NAME	MEM_AGE	MEM_ADDRESS	MEM_PHONE_NUMBER	MEMBERSHIP_ID
103	Abrar	18	Banani, Dhaka	01635453545, 01765789315	24-08
104	Akash	20	Mirpur, Dhaka	01562146879, 01934586721	24-07
105	Junayed	22	Bashundhara, Dhaka	01998977150, 01705008327	24-06
106	Yeasir	22	Gulshan, Dhaka	01316489264, 01732485324	24-05
101	Kabir	21	Uttara, Dhaka	01768352497, 01915482679	24-10
102	Shahriar	27	Farmgate, Dhaka	01993241567, 01701873598	24-09
107	Shuvro	19	Uttara, Dhaka	01648579582, 01912462573	24-04
108	Rasel	26	Badda, Dhaka	01316478594, 01716845972	24-03
109	Nadim	23	Mirpur, Dhaka	01925678942, 01735894254	24-02
110	Abir	25	Banani, Dhaka	01998977150, 01705008327	24-01

3. Payment_Info:

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('A-01', '09-JAN-24', '4500')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('N-02', '15-JAN-24', '4500')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('R-03', '23-JAN-24', '4800')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('S-04', '02-FEB-24', '4500')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('Y-05', '22-FEB-24', '4500')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('J-06', '06-MAR-24', '4300')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('A-07', '27-MAR-24', '4100')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('A-08', '16-APR-24', '4500')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('S-09', '16-APR-24', '5000')

insert into Payment_Info (Payment_ID, Payment_date, amount) values ('K-10', '04-MAY-24', '4500')

PAYMENT_ID	PAYMENT_DATE	AMOUNT
A-01	09-JAN-24	4500
N-02	15-JAN-24	4500
R-03	23-JAN-24	4800
S-04	02-FEB-24	4500
Y-05	22-FEB-24	4500
J-06	06-MAR-24	4300
A-07	27-MAR-24	4100
A-08	16-APR-24	4500
S-09	16-APR-24	5000
K-10	04-MAY-24	4500

4. Members_Payment:

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('101', 'Abir', 25, 'Banani, Dhaka', '01998977150, 01705008327', 'A-01')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('102', 'Nadim', 23, 'Mirpur, Dhaka', '01925678942, 01735894254', 'N-02')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('103', 'Rasel', 26, 'Badda, Dhaka', '01316478594, 01716845972', 'R-03')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('104', 'Shuvro', 19, 'Uttara, Dhaka', '01648579582, 01912462573', 'S-04')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('105', 'Yeasir', 22, 'Gulshan, Dhaka', '01316489264, 01732485324', 'Y-05')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('106', 'Junayed', 22, 'Bashundhara, Dhaka', '01998977150, 01705008327', 'J-06')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('107', 'Akash', 20, 'Mirpur, Dhaka', '01562146879, 01934586721', 'A-07')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('108', 'Abrar', 18, 'Banani, Dhaka', '01635453545, 01765789315', 'A-08')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('109', 'Shahriar', 27, 'Farmgate, Dhaka', '01993241567, 01701873598', 'S-09')

insert into Members_Payment (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, Payment_ID) values ('110', 'Kabir', 21, 'Uttara, Dhaka', '01768352497, 01915482679', 'K-10')

MEM_ID	MEM_NAME	MEM_AGE	MEM_ADDRESS	MEM_PHONE_NUMBER	PAYMENT_ID
103	Rasel	26	Badda, Dhaka	01316478594, 01716845972	R-03
104	Shuvro	19	Uttara, Dhaka	01648579582, 01912462573	S-04
105	Yeasir	22	Gulshan, Dhaka	01316489264, 01732485324	Y-05
106	Junayed	22	Bashundhara, Dhaka	01998977150, 01705008327	J-06
101	Abir	25	Banani, Dhaka	01998977150, 01705008327	A-01
102	Nadim	23	Mirpur, Dhaka	01925678942, 01735894254	N-02
107	Akash	20	Mirpur, Dhaka	01562146879, 01934586721	A-07
108	Abrar	18	Banani, Dhaka	01635453545, 01765789315	A-08
109	Shahriar	27	Farmgate, Dhaka	01993241567, 01701873598	S-09
110	Kabir	21	Uttara, Dhaka	01768352497, 01915482679	K-10

5. Gym:

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-01', 'PowerFlex Gym', 'Fitness', 'Banani, Dhaka', 'Elliptical machine')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-02', 'IronWorks Gym', 'Strength', 'Uttara, Dhaka', 'Rowing machine')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-03', 'BodyForge Gym', 'Cardio', 'Gulshan, Dhaka', 'Smith machine')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-04', 'FlexZone Gym', 'Athletics', 'Banani, Dhaka', 'Stationary bike')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-05', 'CoreFit Gym', 'Fitness', 'Bashundhara, Dhaka', 'Bench press')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-06', 'SweatLab Gym', 'Cardio', 'Mirpur, Dhaka', 'Dumbbells')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-07', 'MuscleMax Gym', 'Athletics', 'Farmgate, Dhaka', 'Cable machine')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-08', 'VitalityX Gym', 'Gymnastics', 'Badda, Dhaka', 'Shoulder machine')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-09', 'PeakPluse Gym', 'Gymnastics', 'Bashundhara, Dhaka', 'Resistance bands')

Insert into Gym (Gym_ID, Gym_name, Gym_type, location, equipment) values ('G-10', 'FitFusion Gym', 'Crossfit', 'Motijheel, Dhaka', 'Dumbbells')

GYM_ID	GYM_NAME	GYM_TYPE	LOCATION	EQUIPMENT
G-01	PowerFlex Gym	Fitness	Banani, Dhaka	Elliptical machine
G-02	IronWorks Gym	Strength	Uttara, Dhaka	Rowing machine
G-03	BodyForge Gym	Cardio	Gulshan, Dhaka	Smith machine
G-04	FlexZone Gym	Athletics	Banani, Dhaka	Stationary bike
G-05	CoreFit Gym	Fitness	Bashundhara, Dhaka	Bench press
G-06	SweatLab Gym	Cardio	Mirpur, Dhaka	Dumbbells
G-07	MuscleMax Gym	Athletics	Farmgate, Dhaka	Cable machine
G-08	VitalityX Gym	Gymnastics	Badda, Dhaka	Shoulder machine
G-09	PeakPluse Gym	Gymnastics	Bashundhara, Dhaka	Resistance bands
G-10	FitFusion Gym	Crossfit	Motijheel, Dhaka	Dumbbells

6. Member_associate_Gym:

CREATE SEQUENCE Member_associate_Gym_MG_ID
INCREMENT BY 1
START WITH 1001
MAXVALUE 99999999
NOCACHE
NOCYCLE;

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '101', 'G-01')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '102', 'G-01')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '103', 'G-01')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '104', 'G-02')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '105', 'G-01')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values
(Member_associate_Gym_MG_ID.nextval, '106', 'G-05')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values (Member_associate_Gym_MG_ID.nextval, '107', 'G-05')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values (Member_associate_Gym_MG_ID.nextval, '108', 'G-07')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values (Member_associate_Gym_MG_ID.nextval, '109', 'G-03')

Insert into Member_associate_Gym (MG_ID, Mem_ID, Gym_ID) values (Member_associate_Gym_MG_ID.nextval, '110', 'G-09')

MG_ID	MEM_ID	GYM_ID
1003	103	G-01
1004	104	G-02
1005	105	G-01
1006	106	G-05
1007	107	G-05
1008	108	G-07
1009	109	G-03
1010	110	G-09
1001	101	G-01
1002	102	G-01

7. Gym_Employee:

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-001', 'Abdul', '01816853071, 01867654981')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-002', 'Hamid', '01725986324, 01632489537')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-003', 'Kashem', '01654987321, 01321654987')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-004', 'Mofiz', '01789564123, 01589321476')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-005', 'Rofique', '01548621973, 01987654321')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-006', 'Mogbul', '01865795325, 01755889944')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-007', 'Ismail', '01564898623, 01321489675')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-008', 'Kamal', '01756489324, 01759846123')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-009', 'Shoikat', '01756843012, 01912356870')

Insert into Gym_Employee (Employee_ID, Employee_name, Employee_phone_number) values ('E-010', 'Rabbani', '01932164852, 01786669537')

EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_PHONE_NUMBER
E-001	Abdul	01816853071, 01867654981
E-002	Hamid	01725986324, 01632489537
E-003	Kashem	01654987321, 01321654987
E-004	Mofiz	01789564123, 01589321476
E-005	Rofique	01548621973, 01987654321
E-006	Mogbul	01865795325, 01755889944
E-007	Ismail	01564898623, 01321489675
E-008	Kamal	01756489324, 01759846123
E-009	Shoikat	01756843012, 01912356870
E-010	Rabbani	01932164852, 01786669537

8. Emp_In_Gym:

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('111', 'G-01', 'E-001')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('112', 'G-02', 'E-002')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('113', 'G-03', 'E-003')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('114', 'G-04', 'E-004')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('115', 'G-05', 'E-005')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('116', 'G-06', 'E-006')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('117', 'G-07', 'E-007')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('118', 'G-08', 'E-008')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('119', 'G-09', 'E-009')

Insert into Emp_In_Gym (GE_ID, Gym_ID, Employee_ID) values ('120', 'G-10', 'E-010')

GE_ID	GYM_ID	EMPLOYEE_ID
111	G-01	E-001
112	G-02	E-002
113	G-03	E-003
114	G-04	E-004
115	G-05	E-005
116	G-06	E-006
117	G-07	E-007
118	G-08	E-008
119	G-09	E-009
120	G-10	E-010

9. Manager:

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-001', 'Farhan', 'A', '01822540711, 01785005407', 'G-01')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-002', 'Anik', 'A', '01829850712, 01675005427', 'G-02')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-003', 'Hridoy', 'B', '01865540731, 01785225487', 'G-03')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-004', 'Tamim', 'A', '01820865711, 01797656499', 'G-04')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-005', 'Prottoy', 'C', '01895843721, 01350867896', 'G-05')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-006', 'Rifat', 'B', '01922540788, 01285005667', 'G-06')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-007', 'Fahad', 'A', '01522540771, 01976005499', 'G-07')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-008', 'Emon', 'A', '01972540722, 01585005777', 'G-08')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-009', 'Shakib', 'C', '01322540771, 01385005223', 'G-09')

Insert into Manager (Manager_ID, Manager_name, Manager_label, Manager_phone_number, Gym_ID) values ('M-010', 'Fahim', 'B', '01622540745, 01885005490', 'G-10')

MANAGER_ID	MANAGER_NAME	MANAGER_LABEL	MANAGER_PHONE_NUMBER	GYM_ID
M-001	Farhan	A	01822540711, 01785005407	G-01
M-002	Anik	A	01829850712, 01675005427	G-02
M-003	Hridoy	B	01865540731, 01785225487	G-03
M-004	Tamim	A	01820865711, 01797656499	G-04
M-005	Prottoy	C	01895843721, 01350867896	G-05
M-006	Rifat	B	01922540788, 01285005667	G-06
M-007	Fahad	A	01522540771, 01976005499	G-07
M-008	Emon	A	01972540722, 01585005777	G-08
M-009	Shakib	C	01322540771, 01385005223	G-09
M-010	Fahim	B	01622540745, 01885005490	G-10

10. Emp_and_Manager:

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME01', 'M-001', 'E-001')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME02', 'M-002', 'E-002')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME03', 'M-003', 'E-003')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME04', 'M-004', 'E-004')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME05', 'M-005', 'E-005')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME06', 'M-006', 'E-006')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME07', 'M-007', 'E-007')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME08', 'M-008', 'E-008')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME09', 'M-009', 'E-009')

Insert into Emp_and_Manager (ME_ID, Manager_ID, Employee_ID) values ('ME10', 'M-010', 'E-010')

ME_ID	MANAGER_ID	EMPLOYEE_ID
ME01	M-001	E-001
ME02	M-002	E-002
ME03	M-003	E-003
ME04	M-004	E-004
ME05	M-005	E-005
ME06	M-006	E-006
ME07	M-007	E-007
ME08	M-008	E-008
ME09	M-009	E-009
ME10	M-010	E-010

11. Trainer_Info:

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-001', 'Atik', 'atik01@gmail.com', '10200')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-002', 'Riyad', 'riyad@gmail.com', '10500')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-003', 'Anik', 'anik77@gmail.com', '10000')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-004', 'Fatin', 'fatin22@gmail.com', '10200')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-005', 'Pritom', 'pritom@gmail.com', '9500')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-006', 'Tanvir', 'tanvir56@gmail.com', '9000')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-007', 'Sami', 'sami107@gmail.com', '11000')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-008', 'Rohan', 'rohan@gmail.com', '12000')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-009', 'Siam', 'siam308@gmail.com', '13000')

Insert into Trainer_Info (Trainer_ID, Trainer_name, Trainer_email, Trainer_salary) values ('T-010', 'Rabbi', 'rabbi190@gmail.com', '11500')

TRAINER_ID	TRAINER_NAME	TRAINER_EMAIL	TRAINER_SALARY
T-001	Atik	atik01@gmail.com	10200
T-002	Riyad	riyad@gmail.com	10500
T-003	Anik	anik77@gmail.com	10000
T-004	Fatin	fatin22@gmail.com	10200
T-005	Pritom	pritom@gmail.com	9500
T-006	Tanvir	tanvir56@gmail.com	9000
T-007	Sami	sami107@gmail.com	11000
T-008	Rohan	rohan@gmail.com	12000
T-009	Siam	siam308@gmail.com	13000
T-010	Rabbi	rabbi190@gmail.com	11500

12. Schedule:

Insert into Schedule (Schedule_ID, "Session", "Activity", Time_duration, Mem_ID) values ('S01', '2:00-4:00 pm', 'Weightlifting', '2 hour', '101')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S02', '2:30-4:30 pm', 'Cardiovascular Exercises', '2 hour', '102')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S03', '2:30-4:00 pm', 'Cardiovascular Exercises', '1.5 hour', '103')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S04', '3:00-5:00 pm', 'Bodyweight Exercises', '2 hour', '104')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S05', '2:00-4:00 pm', 'Functional Training', '2 hour', '105')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S06', '4:00-6:00 pm', 'Weightlifting', '2 hour', '106')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S07', '8:00-10:00 am', 'Bodyweight Exercises', '2 hour', '107')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S08', '7:00-9:00 pm', 'Cardiovascular Exercises', '2 hour', '108')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S09', '8:30-10:30 am', 'Functional Training', '2 hour', '109')

Insert into Schedule (Schedule_ID, "Session", "Activity", time_duration, Mem_ID) values ('S10', '3:00-5:00 pm', 'Functional Training', '2 hour', '110')

SCHEDULE_ID	Session	Activity	TIME_DURATION	MEM_ID
S01	2:00-4:00 pm	Weightlifting	2 hour	101
S02	2:30-4:30 pm	Cardiovascular Exercises	2 hour	102
S03	2:30-4:00 pm	Cardiovascular Exercises	1.5 hour	103
S04	3:00-5:00 pm	Bodyweight Exercises	2 hour	104
S05	2:00-4:00 pm	Functional Training	2 hour	105
S06	4:00-6:00 pm	Weightlifting	2 hour	106
S07	8:00-10:00 am	Bodyweight Exercises	2 hour	107
S08	7:00-9:00 pm	Cardiovascular Exercises	2 hour	108
S09	8:30-10:30 am	Functional Training	2 hour	109
S10	3:00-5:00 pm	Functional Training	2 hour	110

13. Trainer_Period:

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-01', 'T-001', 'S01')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-02', 'T-002', 'S02')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-03', 'T-003', 'S03')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-04', 'T-004', 'S04')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-05', 'T-005', 'S05')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-06', 'T-006', 'S06')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-07', 'T-007', 'S07')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-08', 'T-008', 'S08')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-09', 'T-009', 'S09')

Insert into Trainer_Period (TS_ID, Trainer_ID, Schedule_ID) values ('TS-10', 'T-010', 'S10')

TS_ID	TRAINER_ID	SCHEDULE_ID
TS-01	T-001	S01
TS-02	T-002	S02
TS-03	T-003	S03
TS-04	T-004	S04
TS-05	T-005	S05
TS-06	T-006	S06
TS-07	T-007	S07
TS-08	T-008	S08
TS-09	T-009	S09
TS-10	T-010	S10

14. Mem_and_Trainer:

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-01', '101', 'T-001')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-02', '102', 'T-002')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-03', '103', 'T-003')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-04', '104', 'T-004')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-05', '105', 'T-005')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-06', '106', 'T-006')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-07', '107', 'T-007')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-08', '108', 'T-008')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-09', '109', 'T-009')

Insert into Mem_and_Trainer (MT_ID, Mem_ID, Trainer_ID) values ('MT-10', '110', 'T-010')

MT_ID	MEM_ID	TRAINER_ID
MT-01	101	T-001
MT-02	102	T-002
MT-03	103	T-003
MT-04	104	T-004
MT-05	105	T-005
MT-06	106	T-006
MT-07	107	T-007
MT-08	108	T-008
MT-09	109	T-009
MT-10	110	T-010

15. User_account_Info:

```
CREATE SEQUENCE User_account_Info_User_ID
INCREMENT BY 1
START WITH 1111
MAXVALUE 999999
NOCACHE
NOCYCLE;
Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'abir01', '7894')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'nadim67', '6748')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'rasel25', '9067')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'shuvro22', '8756')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'yeasir56', '8234')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'junayed67', '0348')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'akash52', '8547')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'abrar44', '7725')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'sh808', '6743')

Insert into User_account_Info (User_ID, Username, Password) values (User_account_Info_User_ID.nextval, 'kabir97', '4809')
```

USER_ID	USERNAME	PASSWORD
1111	abir01	7894
1112	nadim67	6748
1113	rasel25	9067
1114	shuvro22	8756
1115	yeasir56	8234
1116	junayed67	0348
1117	akash52	8547
1118	abrar44	7725
1119	sh808	6743
1120	kabir97	4809

16. Mem_User_account:

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('101', 'Abir', 25, 'Banani, Dhaka', '01998977150, 01705008327', '1111')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('102', 'Nadim', 23, 'Mirpur, Dhaka', '01925678942, 01735894254', '1112')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('103', 'Rasel', 26, 'Badda, Dhaka', '01316478594, 01716845972', '1113')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('104', 'Shuvro', 19, 'Uttara, Dhaka', '01648579582, 01912462573', '1114')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('105', 'Yeasir', 22, 'Gulshan, Dhaka', '01316489264, 01732485324', '1115')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('106', 'Junayed', 22, 'Bashundhara, Dhaka', '01998977150, 01705008327', '1116')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('107', 'Akash', 20, 'Mirpur, Dhaka', '01562146879, 01934586721', '1117')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('108', 'Abrar', 18, 'Banani, Dhaka', '01635453545, 01765789315', '1118')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('109', 'Shahriar', 27, 'Farmgate, Dhaka', '01993241567, 01701873598', '1119')

Insert into Mem_User_account (Mem_ID, Mem_name, Mem_age, Mem_address, Mem_phone_number, User_ID) values ('110', 'Kabir', 21, 'Uttara, Dhaka', '01768352497, 01915482679', '1120')

MEM_ID	MEM_NAME	MEM_AGE	MEM_ADDRESS	MEM_PHONE_NUMBER	USER_ID
101	Abir	25	Banani, Dhaka	01998977150, 01705008327	1111
102	Nadim	23	Mirpur, Dhaka	01925678942, 01735894254	1112
103	Rasel	26	Badda, Dhaka	01316478594, 01716845972	1113
104	Shuvro	19	Uttara, Dhaka	01648579582, 01912462573	1114
105	Yeasir	22	Gulshan, Dhaka	01316489264, 01732485324	1115
106	Junayed	22	Bashundhara, Dhaka	01998977150, 01705008327	1116
107	Akash	20	Mirpur, Dhaka	01562146879, 01934586721	1117
108	Abrar	18	Banani, Dhaka	01635453545, 01765789315	1118
109	Shahriar	27	Farmgate, Dhaka	01993241567, 01701873598	1119
110	Kabir	21	Uttara, Dhaka	01768352497, 01915482679	1120

Query Test in DB

Using Conditional Statement:

1. Write a query to display the member ID, name, and age of the member who is under 24.

> SELECT Mem_ID, Mem_name, Mem_age FROM Member WHERE Mem_age < 24

MEM_ID	MEM_NAME	MEM_AGE
103	Abrar	18
104	Akash	20
105	Junayed	22
106	Yeasir	22
101	Kabir	21
107	Shuvro	19
109	Nadim	23

2. Write a query to display the Trainer name salary whose salary between 10,000 and 11,000.

> SELECT Trainer_name, Trainer_Salary from Trainer_Info WHERE Trainer_salary BETWEEN 10000 AND 11000

TRAINER_NAME	TRAINER_SALARY
Atik	10200
Riyad	10500
Anik	10000
Fatin	10200
Sami	11000

3. Write a query to display the Manager ID, name and label whose manager ID is M-007 and label is A

> SELECT Manager_ID, manager_name, manager_label FROM manager WHERE Manager_ID = 'M-007' AND Manager_label = 'A'

MANAGER_ID	MANAGER_NAME	MANAGER_LABEL
M-007	Fahad	A

4. Write a query to display the Member ID and Gym_ID where member do the gym and there gym_ID are G-01 and G-05

> SELECT Mem_ID, Gym_ID FROM Member_associate_Gym WHERE Gym_ID IN ('G-01', 'G-05')

MEM_ID	GYM_ID
103	G-01
105	G-01
106	G-05
107	G-05
101	G-01
102	G-01

5. Write a query to display the Trainer name where the second letter of their name is a.

>SELECT Trainer_name FROM Trainer_Info WHERE Trainer_name LIKE '_a%'

TRAINER_NAME
Fatin
Tanvir
Sami
Rabbi

Using Single row function:

1. Write a query to display the member name whose member ID is 107. (Name should be capital letter)

> SELECT UPPER(Mem_name) as Member_Name FROM Member WHERE Mem_ID = 107

MEMBER_NAME
SHUVRO

2. Write a query to display the all manager ID, name and length of their name.

> SELECT Manager_ID, Manager_Name, LENGTH(Manager_name) as Length_of_name FROM Manager

MANAGER_ID	MANAGER_NAME	LENGTH_OF_NAME
M-001	Farhan	6
M-002	Anik	4
M-003	Hridoy	6
M-004	Tamim	5
M-005	Prottoy	7
M-006	Rifat	5
M-007	Fahad	5
M-008	Emon	4
M-009	Shakib	6
M-010	Fahim	5

3. Write a query to display membership ID, strat date and how many weeks they have passed since they have got membership?

> SELECT Membership_ID, Start_date, ROUND((SYSDATE-Start_date)/7,0) as Weeks FROM Membership

MEMBERSHIP_ID	START_DATE	WEEKS
24-01	10-JAN-24	18
24-02	16-JAN-24	17
24-03	24-JAN-24	16
24-04	03-FEB-24	14
24-05	23-FEB-24	11
24-06	07-MAR-24	10
24-07	28-MAR-24	7
24-08	17-APR-24	4
24-09	17-APR-24	4
24-10	05-MAY-24	1

4. Write a query to display the all membership ID and start date like '24 January 2024'

> SELECT Membership_ID, TO_CHAR (start_date, 'fmDD Month YYYY') as Start_Date from Membership

MEMBERSHIP_ID	START_DATE
24-01	10 January 2024
24-02	16 January 2024
24-03	24 January 2024
24-04	3 February 2024
24-05	23 February 2024
24-06	7 March 2024
24-07	28 March 2024
24-08	17 April 2024
24-09	17 April 2024
24-10	5 May 2024

5. Write a query to display the Trainer's name, salary and bonus where bonus is 10% of their salary.

> SELECT Trainer_name, Trainer_salary, Trainer_salary*0.10 as Bonus FROM Trainer_Info

TRAINER_NAME	TRAINER_SALARY	BONUS
Atik	10200	1020
Riyad	10500	1050
Anik	10000	1000
Fatin	10200	1020
Pritom	9500	950
Tanvir	9000	900
Sami	11000	1100
Rohan	12000	1200
Siam	13000	1300
Rabbi	11500	1150

Using Sub Query:

1. Write a query to display the all information of Trainer where Trainer salary is more than average salary.

> SELECT * FROM Trainer_Info Where Trainer_salary > (SELECT AVG (Trainer_salary) FROM Trainer_Info)

TRAINER_ID	TRAINER_NAME	TRAINER_EMAIL	TRAINER_SALARY
T-007	Sami	sami107@gmail.com	11000
T-008	Rohan	rohan@gmail.com	12000
T-009	Siam	siam308@gmail.com	13000
T-010	Rabbi	rabbi190@gmail.com	11500

2. Write a query to display the payment ID and amount where maximum amount paid by a member.

> SELECT Payment_ID, Amount FROM Payment_Info WHERE Amount = (SELECT MAX(Amount) FROM Payment_Info)

PAYMENT_ID	AMOUNT
S-09	5000

Using Joining:

1. Display the member name, age and start date for all members.

> SELECT m.mem_name, m.mem_age, me.start_date from Member m, Membership me Where m.Membership_ID = me.Membership_ID

MEM_NAME	MEM_AGE	START_DATE
Abrar	18	17-APR-24
Akash	20	28-MAR-24
Junayed	22	07-MAR-24
Yeasir	22	23-FEB-24
Kabir	21	05-MAY-24
Shahriar	27	17-APR-24
Shuvro	19	03-FEB-24
Rasel	26	24-JAN-24
Nadim	23	16-JAN-24
Abir	25	10-JAN-24

2. Display payment ID, Date, ammount and name for all members.

> SELECT p.payment_ID, p.payment_date, p.amount, m.mem_name from Payment_Info p, Members_payment m WHERE p.Payment_ID = m.Payment_ID

PAYMENT_ID	PAYMENT_DATE	AMOUNT	MEM_NAME
R-03	23-JAN-24	4800	Rasel
S-04	02-FEB-24	4500	Shuvro
Y-05	22-FEB-24	4500	Yeasir
J-06	06-MAR-24	4300	Junayed
A-01	09-JAN-24	4500	Abir
N-02	15-JAN-24	4500	Nadim
A-07	27-MAR-24	4100	Akash
A-08	16-APR-24	4500	Abrar
S-09	16-APR-24	5000	Shahriar
K-10	04-MAY-24	4500	Kabir

3. Display member ID and gym name for all members and also display those gym where have no members.

> SELECT m.mem_ID, g.gym_name FROM Member_associate_Gym m, Gym g where m.Gym_ID(+) = g.Gym_ID

MEM_ID	GYM_NAME
105	PowerFlex Gym
103	PowerFlex Gym
102	PowerFlex Gym
101	PowerFlex Gym
104	IronWorks Gym
109	BodyForge Gym
-	FlexZone Gym
106	CoreFit Gym
107	CoreFit Gym
-	SweatLab Gym
108	MuscleMax Gym
-	VitalityX Gym
110	PeakPluse Gym
-	FitFusion Gym

4. Display manager name, label and gym name for all manager

> SELECT m.Manager_name, m.Manager_label, g.Gym_name FROM Manager m, Gym g WHERE m.Gym_ID = g.Gym_ID

MANAGER_NAME	MANAGER_LABEL	GYM_NAME
Farhan	A	PowerFlex Gym
Anik	A	IronWorks Gym
Hridoy	B	BodyForge Gym
Tamim	A	FlexZone Gym
Prottoy	C	CoreFit Gym
Rifat	B	SweatLab Gym
Fahad	A	MuscleMax Gym
Emon	A	VitalityX Gym
Shakib	C	PeakPluse Gym
Fahim	B	FitFusion Gym

Using Multiple row function:

1. Display the trainer's maximum and minimum salary

> SELECT MAX(Trainer_salary), MIN(Trainer_salary) FROM Trainer_Info

MAX(TRAINER_SALARY)	MIN(TRAINER_SALARY)
13000	9000

2. Display the average salary from Trainer Information.

> SELECT AVG(Trainer_salary) FROM Trainer_Info

AVG(TRAINER_SALARY)
10690

3. Display the number of member who do gym in Gym ID G-01.

> SELECT COUNT(*) FROM Member_associate_Gym where Gym_ID = 'G-01'

COUNT(*)
4

VIEW:

Ques: Create a view called Manager_A based Manager ID, name, and label from the Manager table who have label A.

> Create view Manager_A as SELECT Manager_ID, Manager_name, Manager_Label from Manager WHERE Manager_label = 'A'

```
select * from Manager_A
```

Results Explain Describe Saved SQL History

MANAGER_ID	MANAGER_NAME	MANAGER_LABEL
M-001	Farhan	A
M-002	Anik	A
M-004	Tamim	A
M-007	Fahad	A
M-008	Emon	A

Relational Algebra:

1. Display all information for those member whose member ID is 107

$$\left[\begin{array}{l} \text{Member} \\ \sigma \\ \text{Mem_ID} = 107 \end{array}\right]$$

2. Display member name, age and start date for all members.

$$\pi_{\text{m.mem_name, m.mem_age, me.start_date}} \left[\begin{array}{l} (\text{Member} \times \text{Membership}) \\ \sigma \\ \text{m.Membership_Id} = \text{me.Membership_ID} \end{array}\right]$$

3. Display payment ID, Date, ammount and name for all members.

$$\pi_{\text{p.Payment_ID, p.Payment_date, p.Amount, m.mem_name}} \left[\sigma_{\text{p.Payment_Id = m.Payment_ID}} \left[\text{(Payment_Info} \times \text{Members_Payment)} \right] \right]$$

4. Display member name and age whose member ID is 103.

$$\pi_{\text{mem_name, mem_age}} \left[\sigma_{\text{Mem_ID = 103}} \left[\text{Member} \right] \right]$$

5. Write a query to display the member ID, name, and age of the member who is under 24.

$$\pi_{\text{mem_ID, mem_name, mem_age}} \left[\sigma_{\text{Mem_age < 24}} \left[\text{Member} \right] \right]$$

6. Write a query to display the Trainer name salary whose salary between 10,000 and 11,000.

$$\pi_{\text{Trainer_name, Trainer_salary}} \left[\sigma_{\text{Trainer_salary between 10000 and 11000}} \left[\text{(Trainer_Info)} \right] \right]$$

7. Display the trainer's maximum and minimum salary

Trainer_Info

π

MAX (Trainer_salary), MIN (Trainer_salary)

8. Display manager name, label and gym name for all manager.

$$\left[\begin{array}{l}\text{(Manager} \times \text{Gym)} \\ \sigma \\ \text{m.Gym_ID = g.Gym_ID}\end{array}\right]$$

π

m.Manager_name, m.Manager_label, g.Gym_name

9. Write a query to display the Trainer's name, salary and bonus where bonus is 10% of their salary.

Trainer_Info

π

Trainer_name, Trainer_salary, Trainer_salary*0.10 as Bonus

10. Write a query to display the Trainer name where the second letter of their name is a.

>SELECT Trainer_name FROM Trainer_Info WHERE Trainer_name LIKE '_a%'

$$\left[\begin{array}{l}\text{(Trainer_Info)} \\ \sigma \\ \text{Trainer_name LIKE '_a\%'}\end{array}\right]$$

π

Trainer_name

Conclusion

Our gym management system can be used as the database for any gym institution. We tried to incorporate all relatable entity and store their data in the simplest but efficient manner possible. Through this project, we have designed a real world scenario and applied the theoretical knowledge we learned in this Introduction to Database course for a reliable solution. The experience we have gained designing this project will hopefully be useful in developing further database related projects. For the schedule and lack of resources we have developed some limitations in this project, and in future we are very interested adding new features and improving this project.